

Mathematics for Data Science and Machine Learning

Statistics and Probability - Practice Workbook (Volume 3)

Topic 1: Descriptive Statistics

- 1.1** **EASY** Calculate the mean and variance of the dataset [12, 15, 18, 22, 25] (Question 1).
- 1.2** **EASY** Find the median and mode of [3, 3, 5, 8, 8, 8, 10] (Question 2).
- 1.3** **EASY** Determine the range and interquartile range (IQR) of [10, 20, 25, 30, 45, 50, 60] (Question 3).
- 1.4** **EASY** Compute the skewness for a sample with mean 50 and standard deviation 10 (Question 4).
- 1.5** **EASY** Find the coefficient of variation for data with mean 100 and variance 225 (Question 5).
- 1.6** **EASY** Calculate the mean and variance of the dataset [12, 15, 18, 22, 25] (Question 6).
- 1.7** **EASY** Find the median and mode of [3, 3, 5, 8, 8, 8, 10] (Question 7).
- 1.8** **EASY** Determine the range and interquartile range (IQR) of [10, 20, 25, 30, 45, 50, 60] (Question 8).
- 1.9** **INTERMEDIATE** Compute the skewness for a sample with mean 50 and standard deviation 10 (Question 9).
- 1.10** **INTERMEDIATE** Find the coefficient of variation for data with mean 100 and variance 225 (Question 10).
- 1.11** **INTERMEDIATE** Calculate the mean and variance of the dataset [12, 15, 18, 22, 25] (Question 11).
- 1.12** **INTERMEDIATE** Find the median and mode of [3, 3, 5, 8, 8, 8, 10] (Question 12).
- 1.13** **INTERMEDIATE** Determine the range and interquartile range (IQR) of [10, 20, 25, 30, 45, 50, 60] (Question 13).
- 1.14** **INTERMEDIATE** Compute the skewness for a sample with mean 50 and standard deviation 10 (Question 14).
- 1.15** **INTERMEDIATE** Find the coefficient of variation for data with mean 100 and variance 225 (Question 15).
- 1.16** **INTERMEDIATE** Calculate the mean and variance of the dataset [12, 15, 18, 22, 25] (Question 16).
- 1.17** **INTERMEDIATE** Find the median and mode of [3, 3, 5, 8, 8, 8, 10] (Question 17).

- 1.18** **DIFFICULT** Determine the range and interquartile range (IQR) of [10, 20, 25, 30, 45, 50, 60] (Question 18).
- 1.19** **DIFFICULT** Compute the skewness for a sample with mean 50 and standard deviation 10 (Question 19).
- 1.20** **DIFFICULT** Find the coefficient of variation for data with mean 100 and variance 225 (Question 20).
- 1.21** **DIFFICULT** Calculate the mean and variance of the dataset [12, 15, 18, 22, 25] (Question 21).
- 1.22** **DIFFICULT** Find the median and mode of [3, 3, 5, 8, 8, 8, 10] (Question 22).
- 1.23** **DIFFICULT** Determine the range and interquartile range (IQR) of [10, 20, 25, 30, 45, 50, 60] (Question 23).
- 1.24** **DIFFICULT** Compute the skewness for a sample with mean 50 and standard deviation 10 (Question 24).
- 1.25** **DIFFICULT** Find the coefficient of variation for data with mean 100 and variance 225 (Question 25).

Topic 2: Probability Theory

- 2.1** **EASY** Calculate $P(A | B)$ given $P(A)=0.4$, $P(B)=0.5$, $P(B|A)=0.6$ (Question 1).
- 2.2** **EASY** Find the probability of picking 2 red balls from a bag of 5 red and 7 blue balls (Question 2).
- 2.3** **EASY** Determine the probability of event A or B given $P(A)=0.3$, $P(B)=0.4$, $P(A \text{ and } B)=0.1$ (Question 3).
- 2.4** **EASY** If a student passes an exam with prob 0.7, what is prob of failing 2 times in 3 attempts (Question 4).
- 2.5** **EASY** Calculate the probability of drawing an Ace or a Heart from a standard deck (Question 5).
- 2.6** **EASY** Calculate $P(A | B)$ given $P(A)=0.4$, $P(B)=0.5$, $P(B|A)=0.6$ (Question 6).
- 2.7** **EASY** Find the probability of picking 2 red balls from a bag of 5 red and 7 blue balls (Question 7).
- 2.8** **EASY** Determine the probability of event A or B given $P(A)=0.3$, $P(B)=0.4$, $P(A \text{ and } B)=0.1$ (Question 8).
- 2.9** **INTERMEDIATE** If a student passes an exam with prob 0.7, what is prob of failing 2 times in 3 attempts (Question 9).
- 2.10** **INTERMEDIATE** Calculate the probability of drawing an Ace or a Heart from a standard deck (Question 10).
- 2.11** **INTERMEDIATE** Calculate $P(A | B)$ given $P(A)=0.4$, $P(B)=0.5$, $P(B|A)=0.6$ (Question 11).
- 2.12** **INTERMEDIATE** Find the probability of picking 2 red balls from a bag of 5 red and 7 blue balls (Question 12).
- 2.13** **INTERMEDIATE** Determine the probability of event A or B given $P(A)=0.3$, $P(B)=0.4$, $P(A \text{ and } B)=0.1$ (Question 13).
- 2.14** **INTERMEDIATE** If a student passes an exam with prob 0.7, what is prob of failing 2 times in 3 attempts (Question 14).
- 2.15** **INTERMEDIATE** Calculate the probability of drawing an Ace or a Heart from a standard deck (Question 15).
- 2.16** **INTERMEDIATE** Calculate $P(A | B)$ given $P(A)=0.4$, $P(B)=0.5$, $P(B|A)=0.6$ (Question 16).
- 2.17** **INTERMEDIATE** Find the probability of picking 2 red balls from a bag of 5 red and 7 blue balls (Question 17).
- 2.18** **DIFFICULT** Determine the probability of event A or B given $P(A)=0.3$, $P(B)=0.4$, $P(A \text{ and } B)=0.1$ (Question 18).

- 2.19** **DIFFICULT** If a student passes an exam with prob 0.7, what is prob of failing 2 times in 3 attempts (Question 19).
- 2.20** **DIFFICULT** Calculate the probability of drawing an Ace or a Heart from a standard deck (Question 20).
- 2.21** **DIFFICULT** Calculate $P(A | B)$ given $P(A)=0.4$, $P(B)=0.5$, $P(B|A)=0.6$ (Question 21).
- 2.22** **DIFFICULT** Find the probability of picking 2 red balls from a bag of 5 red and 7 blue balls (Question 22).
- 2.23** **DIFFICULT** Determine the probability of event A or B given $P(A)=0.3$, $P(B)=0.4$, $P(A \text{ and } B)=0.1$ (Question 23).
- 2.24** **DIFFICULT** If a student passes an exam with prob 0.7, what is prob of failing 2 times in 3 attempts (Question 24).
- 2.25** **DIFFICULT** Calculate the probability of drawing an Ace or a Heart from a standard deck (Question 25).

Topic 3: Probability Distributions

- 3.1** **EASY** Find the probability $P(X=2)$ for a Binomial distribution with $n=10$, $p=0.3$ (Question 1).
- 3.2** **EASY** Calculate $P(X < 1)$ for a Poisson distribution with mean $\lambda=2$ (Question 2).
- 3.3** **EASY** Find the Z-score for a value $X=110$ in a normal distribution with mean=100 and std=10 (Question 3).
- 3.4** **EASY** Determine the probability $P(-1 < Z < 1)$ in a standard normal distribution (Question 4).
- 3.5** **EASY** Calculate the probability of $X > 5$ for a Uniform distribution $U(0, 10)$ (Question 5).
- 3.6** **EASY** Find the probability $P(X=2)$ for a Binomial distribution with $n=10$, $p=0.3$ (Question 6).
- 3.7** **EASY** Calculate $P(X < 1)$ for a Poisson distribution with mean $\lambda=2$ (Question 7).
- 3.8** **EASY** Find the Z-score for a value $X=110$ in a normal distribution with mean=100 and std=10 (Question 8).
- 3.9** **INTERMEDIATE** Determine the probability $P(-1 < Z < 1)$ in a standard normal distribution (Question 9).
- 3.10** **INTERMEDIATE** Calculate the probability of $X > 5$ for a Uniform distribution $U(0, 10)$ (Question 10).
- 3.11** **INTERMEDIATE** Find the probability $P(X=2)$ for a Binomial distribution with $n=10$, $p=0.3$ (Question 11).
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- 3.13** **INTERMEDIATE** Find the Z-score for a value $X=110$ in a normal distribution with mean=100 and std=10 (Question 13).
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- 3.15** **INTERMEDIATE** Calculate the probability of $X > 5$ for a Uniform distribution $U(0, 10)$ (Question 15).
- 3.16** **INTERMEDIATE** Find the probability $P(X=2)$ for a Binomial distribution with $n=10$, $p=0.3$ (Question 16).
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- 3.20** **DIFFICULT** Calculate the probability of $X > 5$ for a Uniform distribution $U(0, 10)$ (Question 20).
- 3.21** **DIFFICULT** Find the probability $P(X=2)$ for a Binomial distribution with $n=10$, $p=0.3$ (Question 21).
- 3.22** **DIFFICULT** Calculate $P(X < 1)$ for a Poisson distribution with mean $\lambda=2$ (Question 22).
- 3.23** **DIFFICULT** Find the Z-score for a value $X=110$ in a normal distribution with mean=100 and std=10 (Question 23).
- 3.24** **DIFFICULT** Determine the probability $P(-1 < Z < 1)$ in a standard normal distribution (Question 24).
- 3.25** **DIFFICULT** Calculate the probability of $X > 5$ for a Uniform distribution $U(0, 10)$ (Question 25).

Topic 4: Inferential Statistics

- 4.1** **EASY** Perform a Z-test for a sample mean of 52 compared to population mean 50, $n=30$, $\text{std}=5$ (Question 1).
- 4.2** **EASY** Calculate a 95% confidence interval for a sample mean of 100, $n=64$, $\text{std}=16$ (Question 2).
- 4.3** **EASY** Determine the p-value for a T-statistic of 2.1 with 15 degrees of freedom (Question 3).
- 4.4** **EASY** State the Null Hypothesis for testing if the population mean is greater than 100 (Question 4).
- 4.5** **EASY** Perform a Chi-square test of independence for a 2x2 table with values $[[10, 20], [30, 40]]$ (Question 5).
- 4.6** **EASY** Perform a Z-test for a sample mean of 52 compared to population mean 50, $n=30$, $\text{std}=5$ (Question 6).
- 4.7** **EASY** Calculate a 95% confidence interval for a sample mean of 100, $n=64$, $\text{std}=16$ (Question 7).
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- 4.24** **DIFFICULT** State the Null Hypothesis for testing if the population mean is greater than 100 (Question 24).
- 4.25** **DIFFICULT** Perform a Chi-square test of independence for a 2x2 table with values $[[10, 20], [30, 40]]$ (Question 25).